Sr.No. 4608

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26 | Session: Odd MSE October 2025 | Date: 08-10-2025
Sem: 1 | Slot: 10:00am - 11:00am | C. Code: CS413
Exam Type: Written | Block:202 Course Section:1 | Seat No: 2022600003
1_61_461_986_1_1505_48180

Name of the candidate. Dhruv Abhijeet Baljekar

Candidate's UID number: 2 0 2 2 6 0 0 0 0 3

Examination class room number: 2 0 2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BY BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/2025

COURSE NAME: Deep learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

RE/2025/03

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1a)

$$f(w) = \frac{1}{2} \| Xw - y \|^2$$

$$|X^T X - \lambda I| = 0$$

$$\lambda_{max} \cdots \lambda_{min}$$

$$L(\theta) = \frac{1}{n} \sum_{i=1}^{n} (f(w) - y)^2$$

$$\theta_{new} = \theta_{old} - \eta \Delta \theta$$

$$\theta_n = \theta - \eta u$$

$$L(\theta) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{1}{2} \left(\| (Xw - y) \|^2 \right) - y \right)^2$$

$$L(\theta) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{1}{2} (Xw - y)^2 - y \right)^2$$

$$\frac{\partial L(\theta)}{\partial w} = \frac{1}{n} \sum_{i=1}^{n} 2 \left(\frac{1}{2} (Xw - y)^2 - y \right) X (Xw - y)$$

$$\frac{\partial L(\theta)}{\partial w} = \frac{2X}{n} \sum \left(\frac{1}{2} (Xw - y)^2 - y \right) (Xw - y)$$

$$\theta_n - \theta_0 = -\eta u$$

$$\frac{\partial L(\Delta \theta)}{\partial w} = -\eta u = \frac{\partial L(\theta)}{\partial w} \ldots$$

According to PCA.

$$-\eta u = \frac{2X}{n} \sum \left(\frac{1}{2} (Xw - y)^2 - y \right) (Xw - y)$$

Must move opposite

$$\eta < \frac{2X}{un} \sum \left(\frac{1}{2} (Xw - y)^2 - y \right) (Xw - y)$$

proved

$$\boxed{\eta < \frac{2}{\lambda_{max}}}$$

as $\lambda_{max} = \dfrac{1}{\frac{X}{un} \sum \left(\frac{1}{2} (X_w - y)^2 - y \right) (X_w - y)}$

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	Q1.b)	Overfitting happens when the model memorises data instead of learning it. This is why the model's capacity should not be too high. It only takes the input as is, without generalisation or regularisation.

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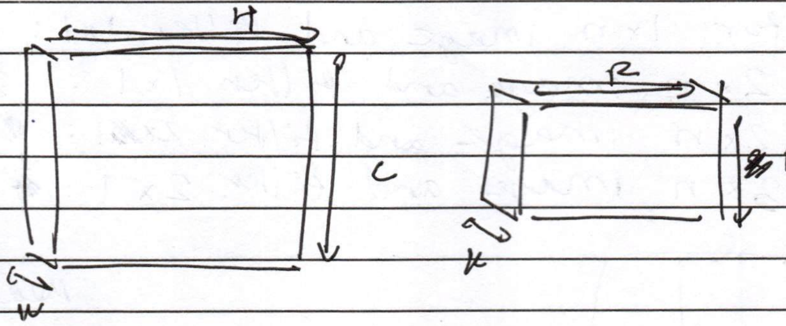
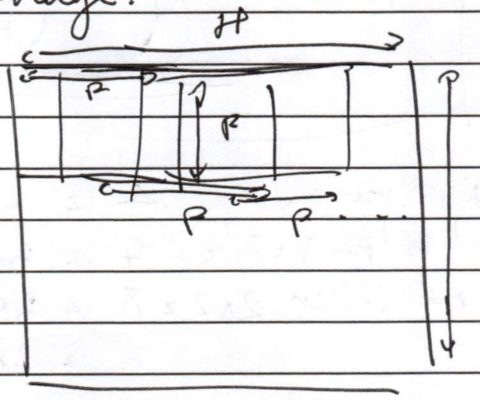
Q2.a)



There will be ~~[illegible]~~ k depth of output image.



F^2 for each filter one

$\therefore F^2 \times$ (no. of times iterated over the image)

$3 \times$ ... $= 9$ how so? $\rightarrow 16$

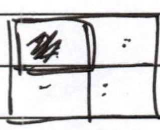
2×2 filter 3×3 image

$F \times H \times C = 3 \times 2 \times$

$F^2((H-F) + (C-F))$

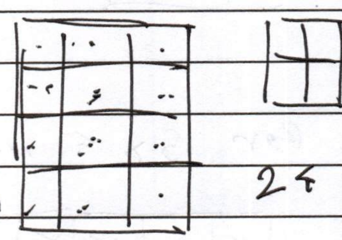
$= 4 \times ((3-2) + (3-2))$

$= 4 \times (2)$



$2 \times 2 = 4$

$1 \times 1 = 1$



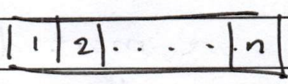
For each square f^2 multiplication

24 multiplication for $f = 2$ & $H = 3$, $C = 4$

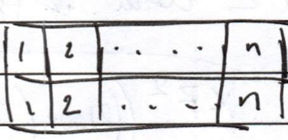
$\therefore f \times H \times C$?

$= 2 \times 3 \times 4 = 24$

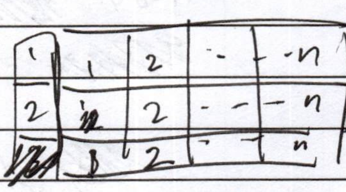
Let us take a $1 \times n$ image with filter 1×1 and then $2 \times n$



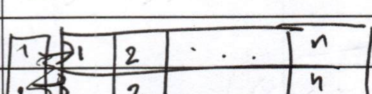
n multiplications



2n multiplications



$2n + 2n = 4n$



1×2 2n multiplications

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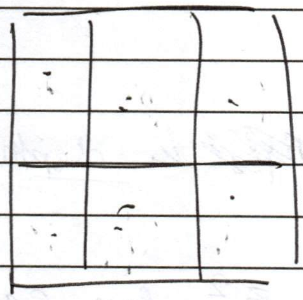
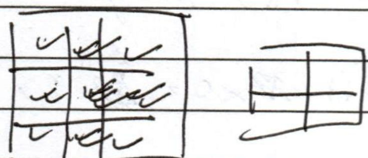
so for $1\times n$ image and filter 1×1 : $n\times1\times1$
For $2\times n$ image and filter 1×1 : $2n\times1\times1$
For $2\times n$ image and filter 2×1 : $n(2\times1)$
For $3\times n$ image and filter 2×1 : $2n(2\times1)$
$\downarrow$
Meaning
$c\times f^2$



$$Ans = F^2\left(\left(floor\left(\frac{C}{F}\right)+1\right)*\left(floor\left(\frac{H}{F}\right)+1\right)\right)$$

For 2×2 image and filter $1\times1 = 4 = 2\times2\times1$
For 3×3 image and filter $1\times1 = 9 = 3\times3\times1$
For 3×2 image and filter $2\times2 = 8 = 2\times2\times(2)$
$= 2\times2\times((3-2)+(2-2))$

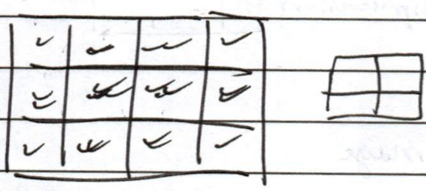
For 3×3 image and filter $2\times2 = 16$



$= 16$

$\therefore$ ~~[illegible]~~ $4\times(4) = 16$
$2\times2\times((3-2)+(3-2))$

For 3×4 image & filter $2\times2 = 24$



~~[illegible]~~ $= 4\times(6)$
~~[illegible]~~

For ~~[illegible]~~ dimension 3×4 we move downwards ~~[illegible]~~ twice if filter is 2×2 and ~~[illegible]~~ sideways 3 times

~~[illegible]~~ $floor\left(\frac{3}{2}\right)+1$ $floor\left(\frac{4}{2}\right)+1$ $\therefore$ $F^2\left(\left(floor\left(\frac{C}{F}\right)+1\right)*\left(floor\left(\frac{H}{F}\right)+1\right)\right)$

For standard : $F^2\left(\left(floor\left(\frac{C}{F}\right)+1\right)* floor\left(\frac{H}{F}\right)+1\right)$

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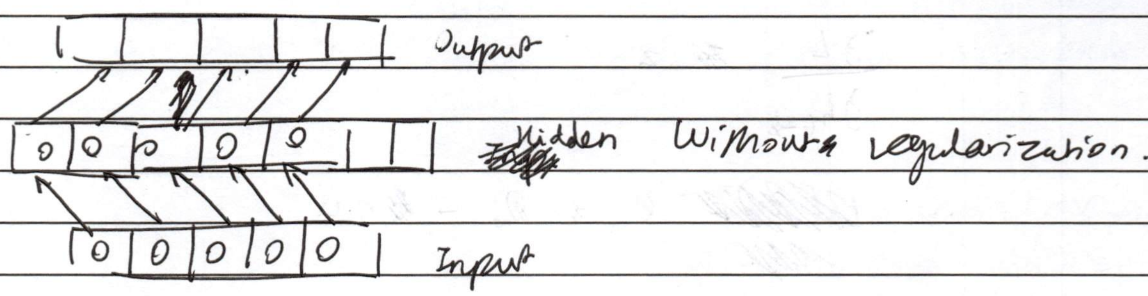
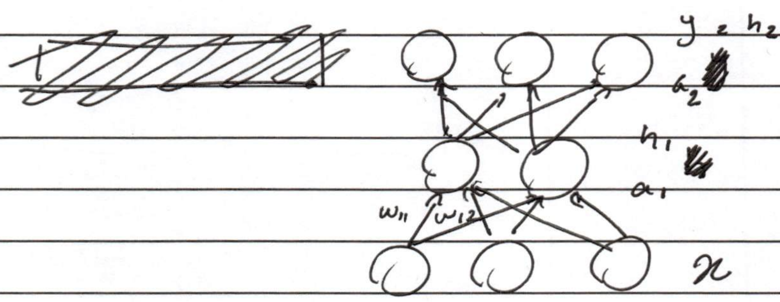
Q2b) An overcomplete autoencoder has larger lattice space than the input, so instead of regularizing it like the undercomplete one does, it simply fits the identity function into the lattice space.





PCA :

$(X^T X - \lambda I) = 0$

$a_1 = w_{11} x_1 + w_{21} x_2 + w_{31} x_3$

$a_i = \sum_{k=0}^{n} w_{ki} x_i$

$h_i = f(a_i)$

$L(\theta) = \frac{1}{n} \sum (h - a)^2)$

$L(\theta) = \frac{1}{n} \sum \left(f(\sum w_{ki} x_i) - \sum w_{ki} x_i \right)$

$\frac{\partial L(\theta)}{\partial w} = \frac{1}{n} \sum \left(x_i f'(\sum w_{ki} x_i) - \sum x_i \right)$

$\frac{\partial L(\theta)}{\partial w} = \frac{1}{n} (PCA)$

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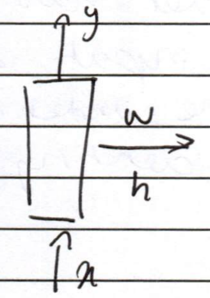
Q.3a)



$h = x_i$

$\frac{\partial L}{\partial h_{t-k}} =$

$\theta_n = \theta_o - \eta u$

The vanishing gradients are solved because the LSTM chooses mathematically what to add, forget and remember, unlike the RNN.

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	Q3b)	GRU has less parameters than LSTM due to the less number of gates, instead of 3, only 2. It improves generalization in the real world by not depending on sequences as much as LSTMs.

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